Fiscal Year:	FY 2026	**Task Last Updated:**	FY 02/04/2026
PI Name:	Massa, Gioia Ph.D.		
Project Title:	Spaceflight Microbiome of a Food Crop Grown Using Different Substrate Moisture Levels		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	NASA Kennedy Space Center		
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City:	Kennedy Space Center	**State:**	FL
Zip Code:	32899-0001	**Congressional District:**	8
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG. App B: Flight and Ground Space Biology Research
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2026 per the PI (Ed., 4/16/26). NOTE: End date changed to 09/30/2025 per D. Freeland/NASA-KSC (Ed., 2/10/25).		
Key Personnel Changes/Previous PI:	Per the PI, Blake Costine was added to the project as a Co-investigator (Ed., 9/2/24). Per the PI, Co-I Jacob Torres left the project in June 2023.		
COI Name (Institution):	Curry, Aaron M.S. (AECOM Management Services, Inc./Kennedy Space Center) Dixit, Anirudha Ph.D. (AECOM Management Services, Inc./Kennedy Space Center) Monje, Oscar Ph.D. (AECOM Management Services, Inc./Kennedy Space Center) Spencer, LaShelle (AECOM Management Services, Inc./Kennedy Space Center) Costine, Blake (Aetos Systems, Inc., Kennedy Space Center)		
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Task Description:	Providing adequate water and oxygen to root zones of crops growing in microgravity has plagued numerous space plant research experiments in a variety of hardware platforms over the years of plant space research. Chronic excess or low water levels, or intermittent watering of plants, leads to alterations in plant growth, and these alterations likely impact the nutritional content and microbiome of those plants. Visible appearances of stress and growth abnormalities have been observed in space crops. In addition, prolonged contact with water, coupled with inhibited airflow, has been shown to lead to opportunistic pathogen development in at least one species on the International Space Station (ISS). Intermittent water stress, such as periodic insufficient water for a rapidly growing plant, can cause loss of turgor and wilting, both on ground and in flight. This type of periodic event may have different impacts on the plant microbiome than prolonged water stress. Our work will apply controlled water stress to assess and quantify changes in plant health and the microbiome in a well-tested food crop, 'Outredgeous' red romaine lettuce, grown under ISS conditions. The Advanced Plant Habitat (APH) hardware has the capability of modulating water flow to crops in a controlled, monitored manner (Massa et al., 2016; Morrow et al., 2016). Thus, we propose to use the APH hardware to intentionally trigger prolonged over- and under-water, and intermittent low-water growth scenarios in this crop, to help to determine how this root zone environment impacts plant health and the plant microbiome. Understanding if the changes in the microbiome of this crop in response to root zone water conditions differ between Earth and space will allow us to better predict the impacts of biophysical variables on future food growth. Results of these studies will help advance fundamental plant and microbial community knowledge and provide strategies and optimum failure modes for a future space food production system that produces high quality fresh vegetables to supplement astronaut diets. References: Massa GD, Wheeler RM, Morrow RC, Levine HG (2016). "Growth chambers on the International Space Station for large plants." Acta Hortic;1134:215-22. https:// Morrow R, Richter R, Tellez G, Monje O, Wheeler R, Massa G, Dufour N, Onate B (2016). "A New Plant Habitat Facility for the ISS." ICES 2016-320. https://
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our work on "Spaceflight Microbiome of a Food Crop Grown Using Different Substrate Moisture Levels" focuses on identifying the impacts of different substrate moisture levels on lettuce crop growth, yield, nutritional composition, and the lettuce microbiome when grown in NASA's Advanced Plant Habitat (APH) plant growth chamber on the International Space Station (ISS). The results of this research will inform space hardware developers of optimal failure modes for water and nutrient delivery and directly translate to Earth-based controlled environment production of lettuce and similar crops in vertical farms and urban plant factories. The capability to grow nutritious, palatable food for crew consumption during spaceflight can potentially provide health-promoting, bioavailable nutrients, enhance the dietary experience, and reduce launch mass as we move toward longer-duration exploration missions. However, studies of edible produce during spaceflight have been limited, leaving a significant knowledge gap in the methods required to grow safe, acceptable, nutritious crops for consumption in microgravity. Providing adequate water and oxygen to the root zones of crops growing in microgravity has plagued numerous space plant research experiments in a variety of hardware platforms over the years of plant space research, and this continues to be a challenge in current ISS experimentation. Given that the complex and often unpredictable behavior of fluids and gasses in collaboration with dynamic biological systems like plants will always be present in a microgravity environment, gaining insight into the impacts of these stresses, both on the plant and the microbiome, will be important. It is expected that the results of these studies will advance fundamental plant and microbial community knowledge and provide strategies and optimal failure modes for a future space food production system that produces high quality fresh vegetables to supplement astronaut diets. To achieve these objectives, first, we have conducted flight-definition testing to determine the moisture levels and watering frequency that best mimic common spaceflight water stress scenarios. Four scenarios for on-orbit testing were identified to best mimic spaceflight scenarios of chronic low water, chronic excess water, intermittent low water to initiate plant wilting and recovery, and optimal water in the substrate. Scenarios were tested with crop growth in a ground analog test bed and these settings were refined and validated in the APH ground hardware. Substrate moisture content was quantified for each proposed condition. Steady state porometry measurements of crop stomatal conductance were assessed to provide a proxy for water stress, but it was decided to not do these assessments in flight due to challenges with available hardware. Next, we plan to grow lettuce on the ISS in the APH with the different moisture scenarios and assess the impact of substrate moisture content on crop yield, physiology, and nutrient content. Plants will be grown on the ISS under the selected four different substrate moisture scenarios to assess impacts to plant growth of chronic and intermittent substrate moisture conditions. Flight plants will be compared with ground controls to assess the impacts of gravity and changes in moisture redistribution in the root zone in microgravity. Plant yield and nutritional content will be assessed. Finally, we will assess the impact of substrate moisture content on the food safety and microbiome of lettuce. We will catalog and characterize the microbiomes of the ISS-grown plants and compare with similar controlled environment-grown plants in the ground hardware. Constant between both ground and ISS will be the endogenous endophytic plant microbiome. Our team will identify changes in response to moisture level and interactions with gravity and spacecraft environmental conditions. Baseline food safety testing will also assess suitability for human consumption.
	To achieve our objectives, first, in earlier reporting periods, we conducted flight-definition testing to determine the moisture levels and watering frequency that best mimic common spaceflight water stress scenarios. Four scenarios for on-orbit testing were identified to best mimic spaceflight scenarios of chronic low water (Drought), chronic excess water (Flood), intermittent low water to initiate plant wilting and recovery (Wilt), and optimal water (Control) in the substrate. Scenarios were tested with crop growth in a ground analog test bed and these settings were refined and validated in the APH ground hardware. Substrate moisture content was quantified for each condition. Spaceflight Microbiome of a Food Crop Grown Using Different Substrate Moisture Levels is research that culminated in an APH experiment on ISS that was designated PH-07. This project has completed preflight definition, with Science Verification Testing (SVT), and Experiment Verification Testing (EVT) conducted in 2024. Between Nov. 2024 and May 2025, we conducted the ISS flight and ground control experiments. We prepared seeds for flight and worked with the payload integration team to integrate seeds and substrates into the APH Science Carrier root trays. Three science carriers and associated growth and harvest hardware were launched to the ISS on CRS-31 in Nov. 2024. Flight tests

Task Progress:	were grown in NASA's Advanced Plant Habitat on the ISS in the Japanese Experiment Module Kibo. Ground controls were conducted at KSC in an ISS Environment Simulator. APH's unique science carrier capability allowed differential moisture control in the root zones of four independent quadrants. ISS plant growth experiments were generally successful, but we had one or two plants in some of the tests that failed to survive the drought or wilt stress. Seed germination and growth rate decreased over time, with the third ISS replicate showing decreased germination and slower growth. The first ground control had very poor germination and was terminated after 8 days because of unacceptable growth. Failure investigation indicated that certain batches of seeds, while showing good germination immediately after sanitization, had an unexpected decline in germination after storage. The batches used in the ground control science carriers had very poor germination, and the batch in the third flight carrier had somewhat poor germination during this failure investigation. The team did a deep dive into the seed sanitation and identified some process failures in controlling the exposure to sanitizing agents which led to some variability in the seed lots. Seed sanitization practices were refined to ensure higher levels of germination. While additional seeds were not able to be sent for the third ISS growth test, additional seed lots were sanitized with tighter process control, and ground controls were replanted. Replanted ground controls showed uniform, excellent germination. Higher water levels were used in early stages of the third ISS replicate to help mitigate germination issues. Plants were grown and monitored for approximately 28 days for each growth test, with photos taken every hour. Screened partitions were added at day 14 to keep shoots in separate quadrants from touching. Moisture treatments were initiated after initial flood filling of the science carrier. Different quadrants in APH were used for each treatment in each growth replicate, however due to limitations on hardware capabilities these treatments were unable to be fully randomized. Control conditions were allowed to dry down from saturation and maintained at an average ~52% volumetric moisture content. Drought treatments were drawn down until the water column was broken and then continued to dry until they reached ~35% moisture content at which point sufficient water was added back daily to keep plants alive. Flood treatments were held at ~2% below total saturation. Wilt treatments were treated like controls for 14 days of growth at which point watering was stopped until plants wilted for 24 hours. Water was added back to reach ~52% % volumetric moisture content, and then stopped again to repeat the wilt stress. The third flight growth test had relatively poor germination and slow growth, likely due to the challenges with seed sanitation mentioned earlier, and this resulted in smaller plants in every treatment. If we look at fresh mass by treatment on average over all harvest periods, we see statistically significant treatment effects between controls and wilt and drought treatments, but not between control and flood, in both flight and ground. While we see similar patterns between treatments in each environment, ground plants are significantly larger than flight plants for control and flood plants and also larger for wilt treated plants. Roots, substrate and wicks were also extracted from all treatments. To extract root zones from wilt and drought treatments, dry substrate had to be rewatered shortly before harvest to enable the root extraction tool to obtain sample cores. Core samples were ejected into bags, and bagged cores were frozen and returned for analysis. When thawed, roots were separated from samples and weighed, and roots, substrates, and wicks were subjected to microbial community sequencing. Flight plant samples are compared with ground controls to assess the impacts of gravity and changes in moisture redistribution in the root zone in microgravity. Nutritional content of edible shoots will focus on measures of potassium, magnesium, iron, total antioxidant capacity and Vitamin C, and these analyses are currently being conducted. These nutrients were selected as they are a primary focus for supplementation of the packaged diet. The food safety of the edible produce was examined, with aerobic plate counts and total yeast and mold counts of culturable microorganisms conducted. Treatments had similar microbial counts, with only the flight drought treatment showing increased variability and slightly higher averages than flight control or ground drought treatments. No specific pathogens were identified in the samples, and the average values coupled with a lack of pathogens indicates that plants from the non-drought treatments would be suitable for human consumption in terms of microbial food safety. We are currently in the process of determining the impact of substrate moisture content on the microbial community associated with the lettuce shoots, roots, substrates, wicks, and water. We have been cataloging and characterizing the microbiomes of the ISS-grown plants and comparing with ground-grown plants and samples. Constant between both ground and ISS is the endogenous endophytic plant microbiome. We are analyzing microbial communities from plant and root-zone samples (water and substrate) using whole-genome metagenomics. Our team will identify changes in response to moisture level and interactions with gravity and spacecraft environmental conditions, with a goal to identify microbial taxa that are shared between, or unique to, the flight and ground treatment conditions. Data analysis is currently underway using Kraken2. Additional work in progress includes image analysis of canopies to determine leaf areas, circadian rhythms, and anthocyanin pigmentation, and to determine the impacts of the stresses on these parameters. Additionally, proteomic analysis of shoot and root tissue is also underway.
Bibliography Type:	Description: (Last Updated: 06/24/2026)
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